BUSA1000 – Introduction to Data, Analytics and People

Unit Coordinator: Dr Hamed Jafarzadeh

Semester 1, 2024

Assessment Task	Case Study/Report
Due date	Refer to iLearn (under 'Case Study/Report' section)
Weight (%)	30% of the total mark
Task description	This is an individual assignment
Submission Method	Via iLearn. Find the submission link under Assessment 2 (Case Study/Report) section
Feedback mechanism(s)	In writing according to marking criteria
Feedback available (anticipated date)	Two to three weeks after the submission due date
Links to Unit Learning Outcomes	ULO2, 3, 4, 5
Word limit:	2,000 works (+/-10%)

INTEGRITY MATTERS

Integrity matters – at home, in your workplace and here at University. As a highly valued member of the Macquarie University Community, you carry great responsibility to uphold the good name of our institution. Our reputation is your reputation and will stay with you for life when your degree opens amazing opportunities for you.

If you are ever unsure whether your actions fall within the guidelines of Academic Integrity, please don't hesitate to reach out. Contact the Academic Literacies Unit or your Tutor/Unit Convenor.

Our purpose at MQBS is to provide inspiring and engaging business education and research that is useful to students throughout their careers and for solving society's biggest problems.

USE OF RESOURCES AND TECHNOLOGIES INCLUDING GENERATIVE ARTIFICIAL INTELLIGENCE

For this assessment, students are NOT permitted to use generative artificial intelligence tools (GAITs), such as ChatGPT, with the exemption of:

- generate preliminary ideas for writing
- edit or improve a working draft of the report
- clarify concepts, theories, ideas, etc., discussed in class

Students are not allowed to use GAITs to:

- Analyse the dataset
- Seek patterns/insights from the data (this should be done with Tableau only)

Any of these actions will constitute a breach of academic integrity and be reported as plagiarism. If students use GAITs to generate initial ideas, they must develop and edit those ideas to a significant extent for them to be submitted as their own work.

Please note that using spelling/grammar checkers (e.g., Microsoft spell-check, Grammarly) is absolutely fine.

Acknowledgment Statement of Use of GAITs:

At the very end of your report, under a clear heading called "Acknowledgment Statement of Use of GAITs", you must select one acknowledgment from the following and include it in your report:

- ☐ I acknowledge that I have **not used** GAITs (e.g., ChatGPT) in drafting and proofreading this assignment.
- ☐ I acknowledge that I have **only used** GAITs (e.g., ChatGPT) in drafting and proofreading this assignment, which is permitted in the assignment instructions.

3 Don'ts in General

- **DON'T** ask a GAIT to complete an assessment for you. This is outsourcing your assessment and is a breach of academic integrity.
- **DON'T** blindly trust GAIT information. GAIT outputs can be completely inaccurate and will often contain fake references. Errors resulting from the uncritical use of GAITs are the students' responsibility and may constitute and be treated as a breach of academic integrity.
- **DON'T** rely on GAITs to replace your own thinking and creativity.

LATE SUBMISSION
A maximum penalty of five percentage points of the total possible marks will be applied per day to late submissions, for up to a maximum of seven calendar days. Tasks that have not been submitted within the maximum number of additional late days will receive a mark of zero, unless otherwise specified in the late penalties section of the Unit Guide. Late submission for a task will only be permitted when specified in the unit guide. This provision does not apply to online exams or other assessment with a time-limit of less than 24 hours. *Where an application for Special Consideration is approved and the outcome is an extension to the due date of a task, submissions that are received after the new due date will be subject to late penalties that are calculated from the new due date. This only applies where the outcome is an extension to the due date – see the Special Consideration Policy (link available in the Unit Guide) for a schedule of all possible outcomes.*

Purpose (the 'Why') of this assignment and the skills to develop

This assignment gives you an opportunity to practice one of the key tasks/responsibilities of business analysts or data analysts: Analysing organizational data and providing insights to managers and decision-makers. You will gain experience in using leading data visualizations technologies (Tableau in this case) which is a highly demanded skill in the market. A great addition to your CV!

Skills in focus for this assessment: Discipline Knowledge, Critical Thinking and Problem Solving, Digital Skills

Client Organisation: Mega Transportation Company (MTC)

Mega Transportation Company (MTC) is a privately held company [1] incorporated in 1975 in Albuquerque, New Mexico, USA. The company is active in the transportation industry and sells a wide range of vehicles ranging from motorbikes and cars to ships, trains, and even airplanes. Their customers are mostly companies and organizations (B2B).

MTC grew rapidly in its first decade. While they started as a local business in the USA, they soon expanded to an international business after a few years with two dozen offices across several countries in five continents. However, in more recent years, they started to see some declines and drops in their revenue and profitability due to competitors' pressure and change of customers' behaviour.

Image source: iStock

The CEO of MTC has recently been to a conference about Business Analytics where several companies shared their fantastic experiences and stories on how analytics help them to achieve business value and enhance competitive advantages. The MTC CEO was amazed to see how data-driven fact-based insights from analytics can improve decision-making in organisations which in turn increase profitability, performance and success. Upon returning from the conference, he placed a high priority on initiating an analytics project in MTC.

[1] It is not a real company! Don't search for it on the Internet! ☺

As a talented student from Macquarie University, you are tasked by MTC to embark them on **Analytics!**

Your Task

As a business analytics specialist, your job is to use **Tableau** to create a visual dashboard for MTC, analyse existing data, and write a report for MTC management providing them with your analysis on sales data.

Download **MTC dataset** (MTC Dataset.xlsx) from iLearn (under Assignment section). This file contains sales information of MTC in 2023. Then use what you have learnt in your BA degree at MQ (along with your self-study and self-training) to create a useful, informative, insightful and rich dashboard for MTC management to use. Think about what diagrams/visualisations would be useful for the managers of MTC (so should be on the dashboard). Then use your dashboard to analyse the MTC sales data and write your report for MTC top management.

The intention of the report is to provide a good and informative summary of data to MTC management and bring the important and interesting highlights to their attention. Think about it this way: *How would you write the report if you were really a staff of MTC and were asked to analyse the sales data and write a report for the top management?* Bring in your judgment, imagination and creativity. Creativity is the spice!

What to include and discuss in your report (and your dashboard) is ultimately your choice. But as a guideline, your report, at least, should address the following points:

1- Screenshot of the ENTIRE dashboard that you have produced with Tableau for MTC management (i.e., one screenshot of the whole dashboard so we can see what you have built). Instead of screenshot, you can also create a PDF or JPG of the dashboard and submit it as an appendix to your report (as a separate file). Make sure to refer to the appendix in an appropriate place in your main report.

   a. At least 9 visualisations (graphs) are required to be included in your dashboard.

   b. One of the visualisations (graphs) should be a trend of sales amount for the entire company over 2023.

c. If the dashboard sheet extent further down your screen (which is likely to happen), take several screenshots (from top and bottom of the sheet) and include them all in your report.

2- Analysis of status of sales from the perspective of products, location of sales (city, country, region), timing of sales, customers, and similar (both in dashboard and the report).

3- Discuss some of the most interesting graphs that you have created. What do they inform?

4- Identify and discuss any interesting insight you can find in the sales information that you would like to bring to the attention of the managers, e.g., any interesting patterns?

5- In addition to above, include a specific dedicated section in your report to answer the below four questions:

- Question 1: How much was the total sales amount in 2023? Per year and quarter.
- Question 2: What was the bestselling country?
- Question 3: What was the bestselling product?
- Question 4: What was the bestselling company, and for which product(s)?

Remember, it is a data-driven report. Everything that you say in the report should be backed-up by data. For example, if you say that product X was sold most, you should provide a screenshot of the relevant graph (from dashboard) in your report to support your finding/statement (including when you are answering the four questions).

Important Notes

- This is an individual assignment. Group work is not allowed, and your report should be your and only your work. The assignment is to be your own analysis and must reflect your own understanding.
- You are expected to deliver a professional piece of business report. Consult this MQ guide on StudyWISE for how to write a good business report: Reports: Reports (mq.edu.au). Also search the internet for finding tips and good examples of business reports. While you have plenty of flexibility, your report is expected to have a good execute summary, introduction, and conclusion.
- No detailed instruction for font type, size, line space, etc! As a University student, however, you are expected to deliver a professional piece of work that is clear, neat and well organised. Don't forget to include page numbers. Figures and tables (if used) should be always numbered and captioned, and you should refer to them at least once in the body of your writing. Using justified text is visually appealing. A good title page, numbering pages, and reasonable page margin are always good practices.
- Clarity of language, layout and general presentation is an assessable aspect of the assignment. The look and feel of the output DOES matter (as it always matters in real-world projects when you produce an output). Always try to impress the reader of your document (which in many cases in real-world projects is your manager or your client) by professionalism, aesthetic, and an appealing look of your work. This is up to you to find out how to do that! Search around the internet, get ideas from good samples, use your own sense of art, etc, etc, etc!
- Remember that your document must look like a professional piece of work. Remember that you are presenting it to your managers of the client companies. Try to impress them by the quality of your work! Managers always care about the appearance and small details.

- MQ's provisions regarding Academic Integrity fully apply. Please refer to the relevant section in the Unit Guide for important information.
- This is a business report assignment. You don't need to use any references from academic literature (good news? ☺). However, if you chose to use references, correct referencing style and formatting for in-text citations and reference list is expected. References (if used) are not counted in the word limit. Any referencing style is acceptable as long as correct, accurate, and consistent.
- Don't change data in the any of the datasets provided for MTC.
- NULL in Excel files means the field is empty/blank.
- You don't have to use all fields in the tables. Some of them might be irrelevant to your purpose. Rely on your judgment.
- This assignment is about using Tableau platform. Using other tools is not acceptable. For example, you cannot analyse the date with Excel or produce diagrams with Excel. Only Tableau.

Submission and deliverables of each part

Submit your report as a Microsoft Word (docx) or PDF file via iLearn. Find the submission link under Assessment 2 Section.

You are allowed to have appendixes to the report. Preferably include them in the single file with the report. If not possible, appendixes could be separate files and submitted along with the main Report file. You should refer to the Appendix(es) in appropriate places in the main body of the report.

Also, Submit the Tableau file of the dashboard (Important Note: The Tableau file should be submitted only in .TWBX format).

Do not forget to include 'Acknowledgment Statement of Use of GAITs' in your report.

Lastly, include a headshot photo of yourself on the cover page of the report. This will help to match names to faces 😊.

Marking criteria

Criteria	Value
Dashboards and Analysis - Designing and developing a useful, informative, insightful and rich dashboard appropriate for managers of MTC - Incorporating adequate, relevant, and meaningful visualisations/diagrams into the dashboard - Good coverage of most important aspects of sales data analysis in dashboard and the report - Thorough analysis of data from different aspects (both in dashboard and the report) - Adequate and well-reasoned discussion/explanation of the (most important) diagrams/visualisations in the report - Reasonable alignment between the dashboard and the report - Reasonable data-driven discussion/insights supported by data and relevant graphs/visualisations - Suitable answer to the four specific questions - A user friendly and interactive dashboard (. TWBX) **Argument Development** - Logical and well-constructed argument/discussion in the report (each paragraph argues one specific point, clear line of argument, good cohesion between paragraphs, clear main thesis statement for each paragraph) - Avoiding irrelevant and off-track information/discussion - Avoiding unsupported statements, claims, and discussions	**75%**
Format and Presentation - Appropriate business report structure - Correct spelling, punctuation and grammar - Clear, logical, concise and professional presentation and writing style - Clear and appropriate language and tone for a professional piece of writing - Readable and good quality screenshots in the report - Adhere to the word count limit - Visually appealing - Adhering to all notes under the “Important Notes” section in the assignment instruction - Inclusion of headshot photo - Inclusion of ‘Acknowledgment Statement of Use of GAITs’ - If applicable, correct referencing style and formatting for in-text citations and reference list.	**25%**
Total	**100%**